KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

HP Elite c640 14 inch G3

Solder-Free Repair Guide



Every field-replaceable repair that needs a screwdriver, not a soldering iron.
HP Elite c640 14 inch G3 Chromebook (Intel Core).

In this guide

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Built from HP's official Maintenance and Service Guide for this model.
Part numbers: verify on partsurfer.hp.com before ordering.

1 Read first

This guide covers every HP Elite c640 14 inch G3 Chromebook repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is in Section 6 and goes to the lead tech or the manufacturer.

How this guide was built. Every screw count, screw size and part number here comes from HP's official Maintenance and Service Guide for the Elite c640 14 inch G3 Chromebook. This guide rewrites HP's procedures step by step for a student bench and adds tests and checks.

Keep the gold strip and silver foil. HP's guide says the gold strip and silver foil inside the bottom cover must stay intact. Don't peel them; they ground the cover.

Bench levels

Level	Who may do it	Typical repairs
LEVEL 1	Any trained student	Software fixes, single keys, bottom cover
LEVEL 2	Student with supervisor sign-off on this repair	Battery, Wi-Fi, touchpad, speakers, USB board, display parts, hinges
LEVEL 3	Lead tech or adult supervisor only	Fan/heat sink, top cover (keyboard) transplant, liquid damage, system board

Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under HP warranty or an accidental-damage plan. A warranty claim may cost less than a part, and opening a unit can affect warranty coverage.
3. ☐ Photograph all four sides of the device before opening it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

Safety rules

- **Power off, unplug, remove any SD card and cables,** then disconnect the battery before any other work inside.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** wear a grounded wrist strap on an ESD mat. Touch boards only by the edges.
- **Ribbon cables:** unlock every ZIF latch before sliding a ribbon out. A locked ribbon tears.
- **Glass:** wear safety glasses when removing a cracked screen.
- **Before power-on:** check for loose screws inside. Shake gently and listen. A loose screw can short the board.

Tools and supplies

Item	Use	Notes
Phillips #1 (magnetic) and #0	All screws	HP lists a magnetic Phillips P1
Tweezers	Stretch-release panel tape	Pull slowly and flat
Plastic spudger, opening picks, plastic tweezers	Clips, latches, ribbons	Only plastic touches connectors
ESD mat + wrist strap	Static protection	
Magnetic screw mat	Keeping screws in order	Screws are often different lengths
90%+ isopropyl alcohol + soft brush	Cleaning contacts, spill residue	
Kapton tape	Re-taping cables and connectors	
Recovery USB or SD card, 8 GB+	ChromeOS recovery	Chromebook Recovery Utility
Known-good USB-C charger	Testing power and charging	
Safety glasses	Cracked-screen removal	
Fire-safe battery bin or sand bucket	Damaged batteries	Required at every bench

2 Know the machine

Specs that matter for repair

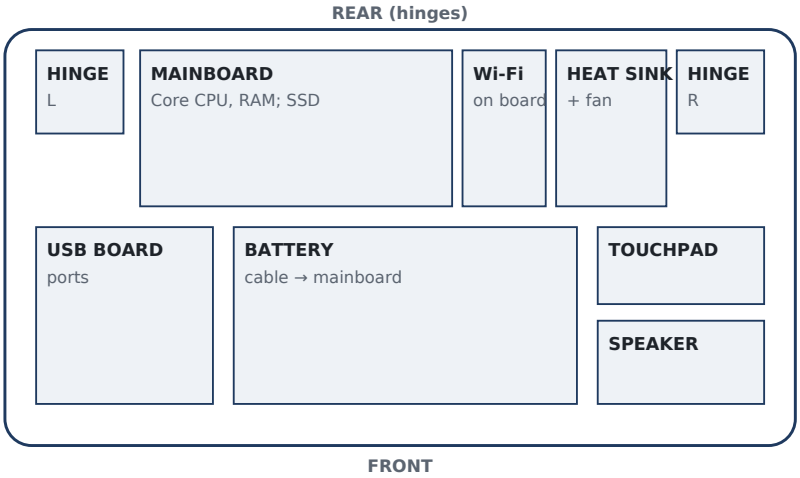
Item	Elite c640 G3	Why it matters
Processor	Intel Core (11th gen), soldered	System board level
Memory / storage	Soldered RAM; SSD on the system board	Move the SSD to a new board
Display	14" FHD, glued bezel, tape-mounted panel	Bezel and panel need new adhesive
Battery	HP M12585-005	5 screws
Access	Bottom cover: 6 screws	Pry, then lift
Wireless	No separate Wi-Fi spare in HP's guide	Treat as system-board level
Cooling	Fan + heat sink	Clean and re-paste every time

How it comes apart

A screwed-on bottom cover comes off first. Under it: battery, system board, fan, heat sink, USB board, speakers and touchpad. The keyboard is built into the top cover, which is the last part left after everything else is removed.

Inside the base

With the cover off you see the parts below. It's a rough map; correct it on your first teardown.



ILLUSTRATIVE — block positions are approximate. Confirm on your first teardown.

Elite c640 G3 parts list

HP spare part numbers from the official Maintenance and Service Guide. Confirm on partsurfer.hp.com by serial number before ordering.

Part	Part number	Repair
Bottom cover	N12891-001	R2
Battery	M12585-005	R3
Touchpad / cable	N13682-001 / N14811-001	R5
Speakers (pair)	N12919-001	R6
USB board / cable	N13679-001 / N13681-001	R7
Fan / heat sink	N12918-001 / N13683-001	R8
Display bezel / adhesive kit	N12925-001 / M17753-001	R10
FHD panel / tape kit	N13685-001 / N13937-001	R11
Display cable	N12926-001	R12
Webcam module / cable	N12929-001 / N10434-001	R13
Hinges / hinge cover	N12927-001 / N12928-001	R14
Fingerprint reader / board	N14814-001 / N12915-001	R15
Top cover with keyboard	N14927-xxx (by country)	R15
System board	N13676-001 (example)	R17

3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through "try first" before ordering parts.

Symptom	Try first	Then	Part likely needed
Won't turn on, no lights	Known-good charger 30 min → S1	R4 → R3	Battery, or system board
Not charging	Other USB-C port and charger → S1	R3 → Sec. 6 if a port is dead	Battery or system board
Battery drains fast / low health in S2	S2 battery test	R3	Battery
Swollen battery / bulging case	STOP — Level 3	R3 by lead tech	Battery
Black screen, power light on	External display via USB-C; S1	R4 → R12 → R11	Display cable or panel
Flicker or lines that change with lid angle	S1	R12	Display cable
Cracked or blotchy screen	—	R11	LCD panel
Bezel loose or broken	Reseat clips	R10	Bezel
Lid floppy, hinge cracked	Check hinge screws	R14	Hinge(s)
One key missing or broken	—	R1	Keycap
Several keys or a row dead	S2 keyboard test; S1; S4	R4 → R15	Keyboard / top cover
Keys type wrong symbols	S4	S4	None
Touchpad dead or erratic	S1; check touchpad settings	R4 → R5	Touchpad or its cable
USB-A / SD card / audio dead	Try another device in the port; S1	R7	I/O board or its cable
Camera not detected	Privacy shutter open? S1	R13	Camera module
No sound (headphones work)	Volume, output; S1; S3	R6	Speakers
Runs hot, fan loud, shuts down	Check vents for lint; S2	R8	Fan or heat sink
Liquid spill	Power off NOW, unplug	R16	Depends on damage
"ChromeOS is missing or damaged"	S3 recovery	R17	System board (storage)
Cracked bottom or lid shell	—	R2	Cover

4 Software repairs

No tools needed. Always do S1 before opening a device.

S1 Hard reset (EC reset) LEVEL 1

Fixes	Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad
Time	2 minutes
Tools	Known-good USB-C charger
Data loss	None

A hard reset restarts the embedded controller (power, keyboard and charging chip). It erases nothing.

Procedure

1. ☐ Unplug the charger and any USB devices. Remove any SD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Refresh** (circular arrow), then tap **Power**. Release Refresh when it starts.
4. ☐ If nothing happens: plug in a known-good charger, wait 30 minutes, and repeat with the charger connected. Try the other USB-C port too.

Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

Tip. After a battery has been disconnected, plug in a charger before deciding the device is dead. Many Chromebooks won't wake on the power button alone.

S2 Built-in hardware diagnostics LEVEL 1

Fixes	Confirms the fault before ordering a part: battery, CPU, memory, keyboard, touchpad, Wi-Fi
Time	5–15 minutes
Tools	None
Data loss	None

Procedure

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the Launcher, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key; test the touchpad and touchscreen.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

S3 Powerwash and USB recovery — LEVEL 1

Fixes	Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", Wi-Fi missing after an update
Time	Powerwash 10 min; recovery 20–40 min
Tools	Recovery USB or SD card (8 GB+)
Data loss	**Yes.** Erases all local files (Downloads).

District-managed devices. Enrolled devices should force re-enrollment on their first Wi-Fi connection after a wipe. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi reaches enrollment. Never enter Developer Mode on a district device.

Procedure

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates.
2. ☐ **Powerwash:** sign out, press Ctrl + Alt + Shift + R, choose Restart, then Powerwash. (May be blocked by policy; use recovery instead.)
3. ☐ **Make recovery media:** on a working computer, open the Chromebook Recovery Utility, choose the **HP Elite c640 14 inch G3 Chromebook** from its list (or search the model), insert an 8 GB+ drive, and click Create now.
4. ☐ Hold **Esc + Refresh**, tap **Power**, and release when the recovery screen appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm it re-enrolls to the KPS domain.

Verify before closing the ticket

- Boots to the enrollment or KPS sign-in screen.
- Shows as enrolled in the Admin console.
- Recovery can't write to storage: storage has failed. That's a system board repair or warranty claim.

S4 Keyboard layout and international keyboard fix — LEVEL 1

Fixes	Keys type the wrong symbols (@ and " swapped, accents appear, Z and Y swapped); a replacement keyboard with a different layout
Time	5 minutes
Tools	None
Data loss	None

ChromeOS can have several keyboard layouts installed. Students often switch layouts by accident with **Ctrl + Shift + Space**, or have **US International** on, which turns ' and " into accent keys. The physical keyboard is fine.

Procedure

1. ☐ Look at the bottom-right shelf. A layout code (US, INTL, UK, ES...) means more than one layout is installed.
2. ☐ Open **Settings > Device > Keyboard and inputs > Input** (on older versions: Settings > Advanced > Languages and inputs > Inputs).
3. ☐ Set **English (US)** as the active input. Remove any layouts the student doesn't need, especially US International, unless a teacher asked for it.
4. ☐ Test in a text box: Shift + 2 should type @ and Shift + ' should type ".
5. ☐ If a **replacement keyboard** has a different physical layout (UK keyboard on a US unit, for example), order the correct US part. Changing the software layout won't fix printed key labels.
6. ☐ If the wrong layout keeps coming back for everyone who signs in, it's set by district policy. Tell the lead tech; don't change it on the device.

Verify before closing the ticket

- Every key types its printed character in the S2 keyboard tester and in a text box.
- Only the needed layouts remain installed.

5 Hardware repairs

Most repairs start with the bottom cover (R2) and disconnecting the battery (R3). Screw counts and sizes come from HP's guide.

Golden rules. Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

R1 Single keycap replacement — LEVEL 1

Fixes	Missing, cracked or loose key
Part	Individual key (ask the manufacturer or harvest from a scrapped keyboard)
Time	5 minutes
Tools	Plastic pick, tweezers

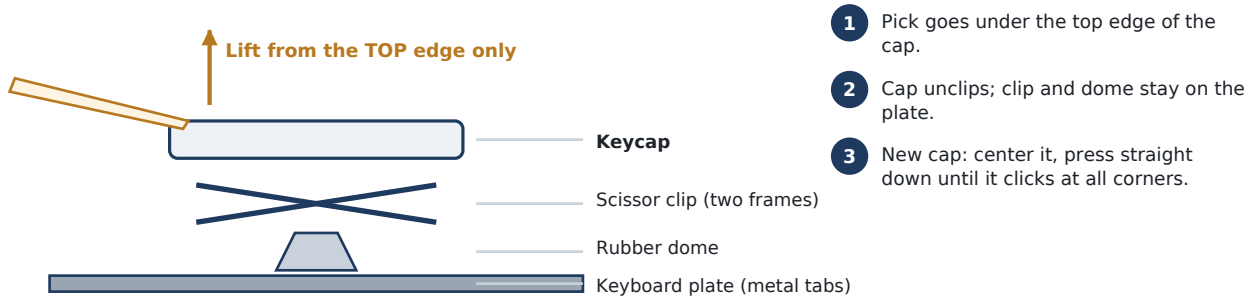


Illustration: side view of one key. Not to scale.

Procedure

1. ☐ Power off. Open the lid flat on the mat.
2. ☐ Slide a plastic pick under the TOP edge of the damaged key and lift until it unclips.
3. ☐ Check the scissor clip and rubber dome under it. A broken clip must be replaced too.
4. ☐ Fit a clip: hook its pins under the metal tabs on the keyboard plate, then press flat.
5. ☐ Center the new key over the clip and press straight down until it clicks at every corner.
6. ☐ Wide keys (Space, Shift, Enter): seat the metal stabilizer wire in its holders first.

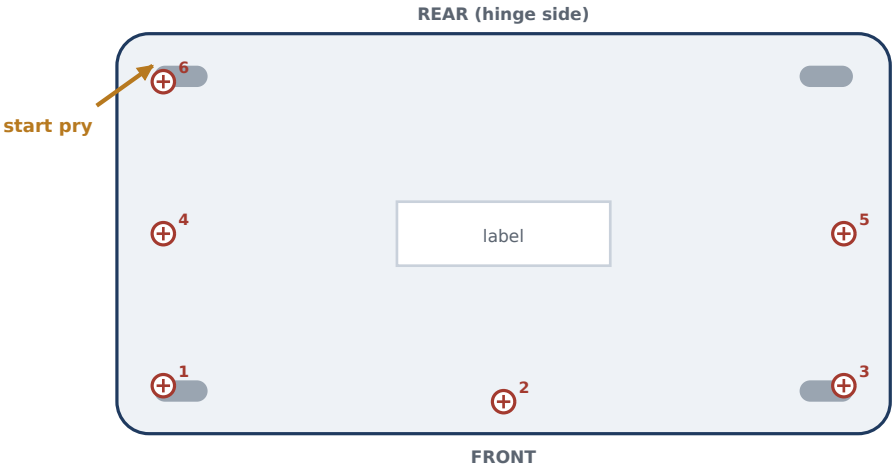
Verify before closing the ticket

- Key clicks evenly and springs back.
- Key registers in the S2 keyboard tester.

R2 Base cover (bottom cover) removal

LEVEL 1

Fixes	Cracked or missing bottom cover; the first step for most internal repairs
Part	Bottom cover N12891-001
Time	10 minutes
Tools	Phillips driver, plastic picks, screw mat



1. Remove the 6 bottom screws.
2. Lay screws on the mat in this same pattern.
3. Pry from the corner shown, then work around the edge.
4. Disconnect the battery before anything else.

Illustration: underside. Screw count from HP's guide; positions illustrative.

Procedure

1. ☐ Shut down fully, unplug, remove any SD card.
2. ☐ Close the lid and turn the computer upside down on the mat, hinges away from you.
3. ☐ Remove the 6 screws: 4 Phillips M2.5×6.0 along the top (hinge) edge and 2 Phillips M2.0×4.0 along the front edge.
4. ☐ Release the cover with a plastic pick at the rear edge, then work around the sides. Lift it off.
5. ☐ **Disconnect the battery now** (battery repair, first steps) before anything else.
6. ☐ To refit: line up the cover, press around the edges until every clip snaps, then refit every screw. Snug only. Check the gold strip and silver foil are still in place.

Verify before closing the ticket

- No gaps around the seam; device sits flat.
- Every screw refitted, none left over.

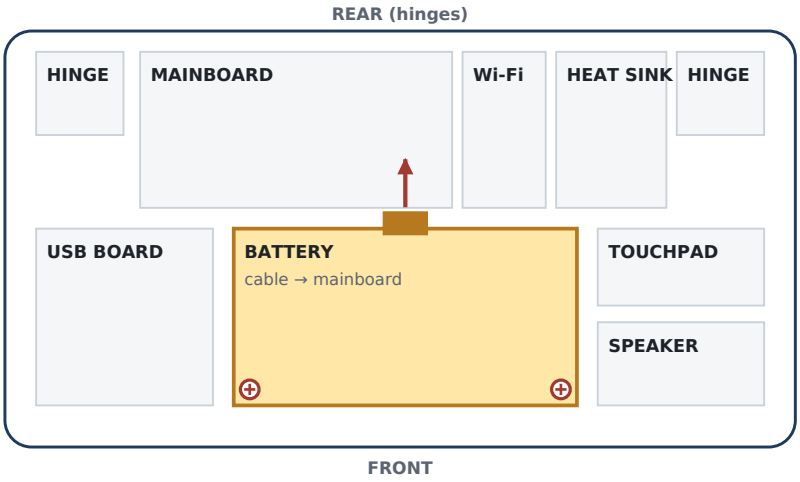
First-teardown notes — fill in the first time you do this repair

Which holes take the longer screws? _____

R3 Battery replacement

LEVEL 2

Fixes	Won't hold charge, low health in S2, swollen battery
Part	Battery M12585-005
Time	15–20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap



1. Unplug the battery cable from the mainboard first (red arrow).
2. Pry at the plug body, never the wires.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

Illustration: base interior, battery highlighted. Screw positions illustrative.

Swollen battery = Level 3. Don't press or pry. The lead tech handles it; the old cell goes straight to the fire-safe bin.

Procedure

1. ☐ Remove the bottom cover.
2. ☐ Lift any tape over the battery connector and disconnect the battery cable from the system board.
3. ☐ Remove the 5 Phillips M2.0×4.0 screws.
4. ☐ Lift the battery out. Never bend it.
5. ☐ Fit the new battery, refit the screws, reconnect the cable.
6. ☐ Refit the bottom cover. Plug in the charger before powering on.

Verify before closing the ticket

- Charging light comes on; battery percentage shows.
- S2 battery health near 100% on a new battery.
- Old battery goes to battery recycling, never the trash.

HP battery notes. HP says to use only HP-supplied batteries. If a battery is swollen or stuck, stop and contact HP support rather than forcing it.

R4 Reseating internal connectors

LEVEL 2

Fixes	Dead keyboard, touchpad or speakers; black screen; dead ports; no power after a drop
Part	None, unless a cable is damaged
Time	15–25 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol

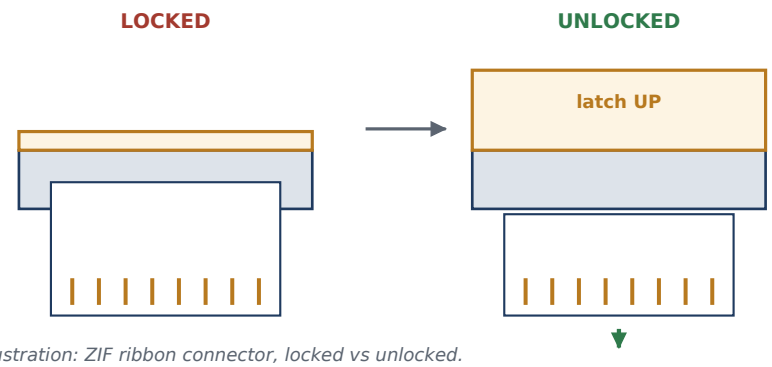


Illustration: ZIF ribbon connector, locked vs unlocked.

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Connector	Type	How to release
Battery cable	Board-to-wire	Lift tape; pry at the connector body
Keyboard ribbon	ZIF ribbon	Unlock the latch; slide out
Touchpad ribbon	ZIF ribbon	Unlock the latch; slide out
Display (eDP) cable	Locking connector	Lift tape and the locking bar, pull straight
USB board cable	ZIF ribbon(s)	Unlock both ends first
Speaker cable	Board-to-wire	Pry at the plug body
Fan cable	Board-to-wire	Pry at the plug body
Fingerprint reader cable	ZIF ribbon	Unlock the latch; slide out

Procedure

1. ☐ Open the device and disconnect the battery.
2. ☐ Hold Power for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ For each connector related to the symptom: photograph it, unlock and release it, and inspect it for bent pins, burn marks or creases.
4. ☐ Clean dirty contacts with a brush dampened in isopropyl alcohol. Let them dry 5 minutes.
5. ☐ Reconnect each one. Ribbons must go in straight and all the way before you close the latch.
6. ☐ Reconnect the battery last. Plug in and test **before** closing up.

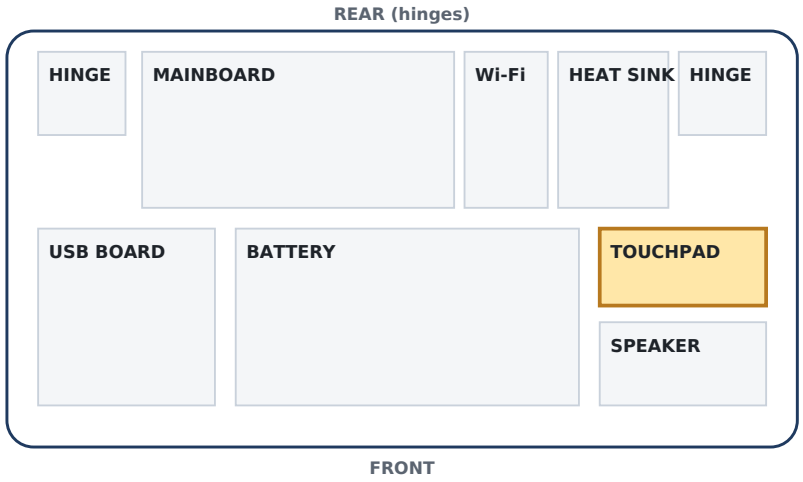
Verify before closing the ticket

- Symptom gone in a full test, not just at boot.
- Creased or torn ribbons replaced.

R5 Touchpad and touchpad cable replacement

LEVEL 2

Fixes	Touchpad dead, erratic, cracked or won't click (keyboard works)
Part	Touchpad N13682-001; touchpad cable N14811-001
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape



1. Battery disconnected first.
2. Unlock the ZIF latch before sliding the ribbon out.
3. Lift tape or Mylar; remove the screws.
4. Conductive tape must touch the pad on reassembly.

Illustration: base interior, touchpad highlighted. Position illustrative.

Check first: touchpad turned on in Settings, S1 done, and the cable reseated.

Procedure

1. ☐ Remove the bottom cover and the battery.
2. ☐ Lift the Mylar over the touchpad connector and disconnect the touchpad cable (ZIF).
3. ☐ Remove the 3 Phillips M2.0×2.0 screws.
4. ☐ Lift the touchpad out. The cable is a separate part (N14811-001): transfer or replace it.
5. ☐ Fit the new pad, refit the screws and Mylar, reconnect.

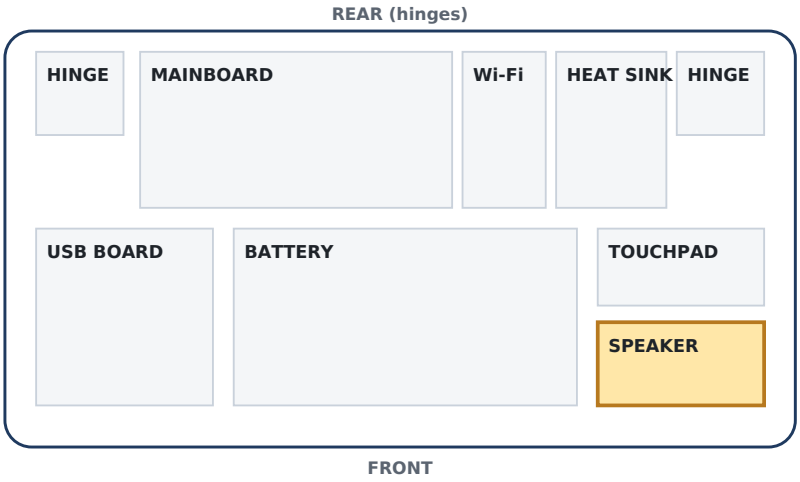
Verify before closing the ticket

- Cursor moves, clicks and two-finger scrolls.
- Click works at all four corners.
- Passes the S2 touchpad test.

R6 Speaker replacement

LEVEL 2

Fixes	No sound or crackling from the speakers, with headphones working
Part	Speakers N12919-001 (pair)
Time	20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers



1. Battery disconnected.
2. Unplug the speaker cable at the board.
3. Unroute the cable, noting each clip.
4. Keep the rubber isolators: they stop rattle.

Illustration: base interior, speaker highlighted. Position illustrative.

Check first: if headphones also have no sound, the fault is software or the system board, not the speakers.

Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the speaker cable from the system board and free it from its routing clips.
3. ☐ Remove the 4 Phillips M2.0×5.0 screws (2 per speaker). Keep the rubber washers.
4. ☐ Lift both speakers out together.
5. ☐ Fit the new pair with the washers, route the cable and reconnect.

Verify before closing the ticket

- Clear sound from both sides with no rattle at full volume.
- Headphone jack still works.

R7 I/O (USB) daughterboard replacement

LEVEL 2

Fixes	Dead USB-A, audio jack with the system board working
Part	USB board N13679-001; cable N13681-001
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers

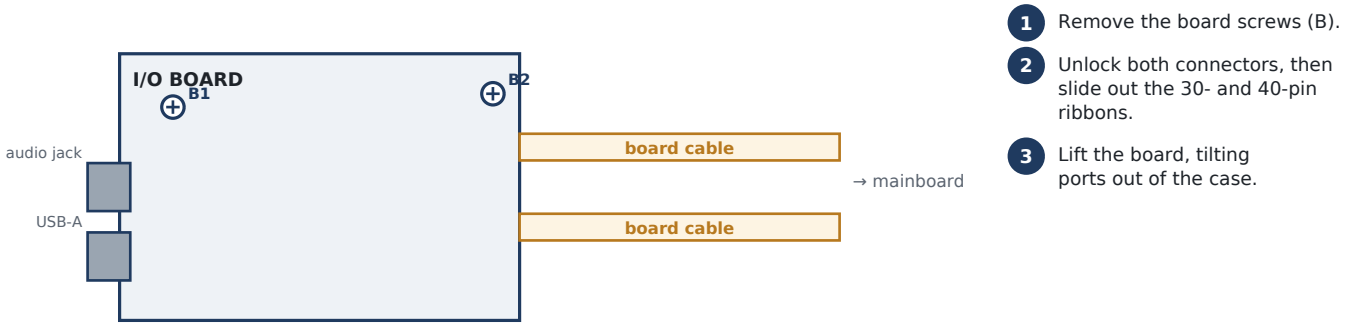


Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.

Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Remove the bracket (2 Phillips M2.0×3.0) and the cable clip (1 Phillips M2.0×2.5).
3. ☐ Disconnect the USB board cable at both ends.
4. ☐ Remove the 3 Phillips M2.0×2.5 screws and lift the board out.
5. ☐ Fit the new board and its cable (N13681-001), refit the clip and bracket.

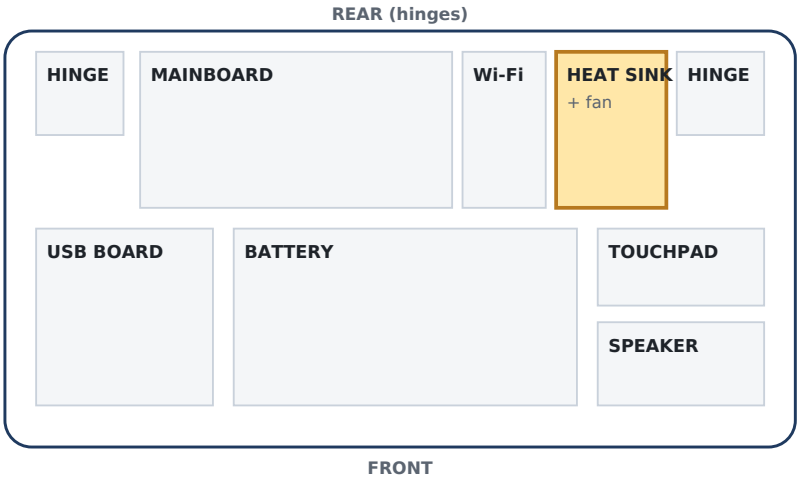
Verify before closing the ticket

- Every port on the board works (flash drive, charger, SD card, headphones as fitted).
- Bracket and all screws refitted.

R8 Fan and heat sink replacement

LEVEL 3

Fixes	Overheating, thermal shutdowns, loud or rattling fan
Part	Fan N12918-001; heat sink N13683-001
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol, lint-free cloth



1. Battery disconnected.
2. Loosen the screws in the numbered order printed on the heat sink.
3. Clean old thermal paste with alcohol.
4. New thermal material, then tighten in order.

Illustration: base interior, cooling highlighted. Position illustrative.

Thermal material. Clean old thermal material off the processor and heat sink every time, and fit new material. Never run the board without the heat sink.

Procedure

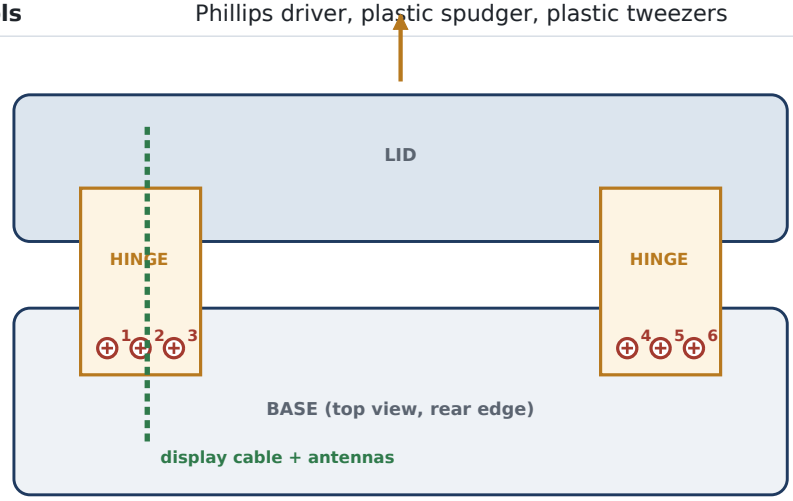
1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the fan cable from the system board and remove the 3 Phillips M2.0×3.0 screws holding the fan. Lift it out.
3. ☐ Loosen the 4 Phillips M2.0×3.0 screws in the order printed on the heat sink, and lift it off.
4. ☐ Clean all old thermal paste and pads off the processor and the heat sink.
5. ☐ Apply new thermal material (included with the heat sink and board kits), refit the heat sink in order, then the fan.

Verify before closing the ticket

- Device runs a 10-minute video without shutting down.
- Fan (if fitted) spins up under load and is quiet at idle.

R9 Display assembly removal (separating lid from base) LEVEL 2

Fixes	Needed for hinge, lid and some board repairs; replacing a whole display assembly
Part	Display spared by subcomponent (bezel, panel, cable, camera, hinges, back cover)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers



1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Remove the display-to-base hinge screws (count them).
4. Support the lid, then lift it away from the base.

Illustration: hinge area seen from above with the D cover off. Positions approximate.

Procedure

1. ☐ Remove the bottom cover and disconnect the battery. Remove the heat sink first on this model.
2. ☐ Disconnect the display cable and both antennas from the system board; free them from their routing.
3. ☐ Remove the 4 Phillips M2.5×4.5 hinge screws.
4. ☐ Open the display to 90°, support it, and lift it away from the base.
5. ☐ Refit in reverse; route the antennas and display cable back through their clips.

Verify before closing the ticket

- Display, camera and Wi-Fi all work after reassembly.
- Lid opens through its full range without pinching cables.

R10 Display bezel replacement

LEVEL 2

Fixes	Cracked, loose or missing screen bezel; the first step for panel, camera and display-cable repairs
Part	Display bezel N12925-001; bezel adhesive kit M17753-001
Time	10 minutes
Tools	Plastic picks, Phillips driver

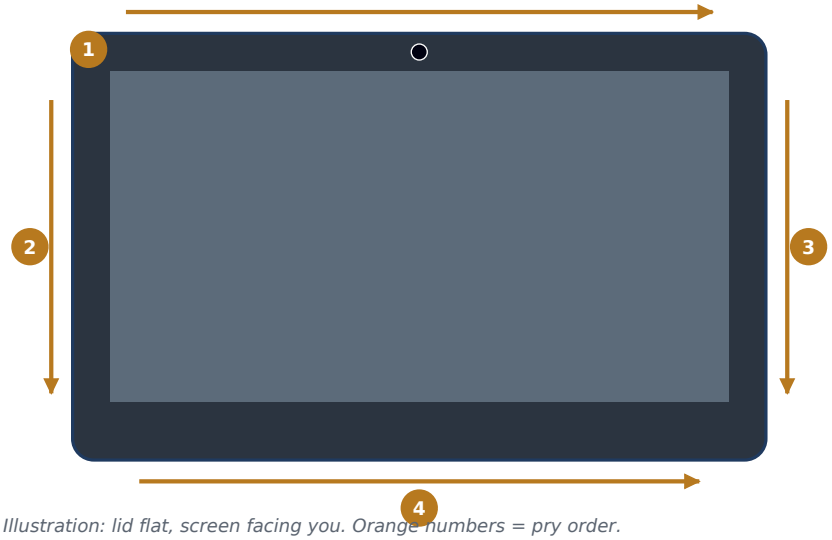


Illustration: lid flat, screen facing you. Orange numbers = pry order.

1. This bezel is glued as well as clipped.
2. Warm the edge gently (hair dryer on low) if it resists.
3. Slide a thin pick along the adhesive; don't bend the bezel.
4. Fit new adhesive strips on reassembly.
5. Don't pull on the camera window.

Procedure

1. ☐ Open the lid flat.
2. ☐ The bezel is held with adhesive on all four sides plus hooks on the top and bottom. Warm it gently if it resists.
3. ☐ Work a thin pick under the edge and release it side by side, then unhook the top and bottom.
4. ☐ Clean off old adhesive and fit the adhesive kit (M17753-001) before refitting the bezel.

Verify before closing the ticket

- Bezel sits flush all round.
- Camera window lined up; privacy shutter slides freely.
- Lid closes fully.

R11 LCD panel replacement

LEVEL 2

Fixes	Cracked screen, dead pixels, blotches, black screen not fixed by the display cable
Part	FHD panel N13685-001 (match your label); panel tape kit N13937-001
Time	25–35 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, safety glasses, Kapton tape

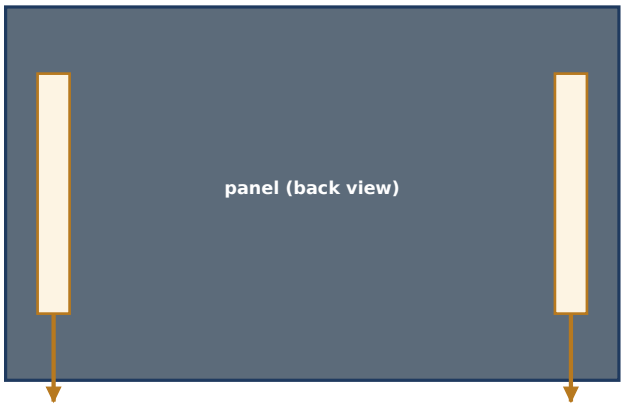


Illustration: stretch-release panel tape. Tape position illustrative.

1. Bezel off first. The panel has no screws.
2. Find the stretch-release tape tabs under the left and right edges.
3. Pull each tab slowly, almost flat to the screen, with tweezers. It may take several pulls.
4. Never pry the panel off over unreleased tape; it cracks.
5. Fit new tape strips on reassembly.

Cracked glass. Wear safety glasses. Tape across a shattered panel before handling it.

Test first: if an external USB-C display works, the system board is fine and the fault is the panel or its cable.

Procedure

1. ☐ Remove the bezel.
2. ☐ Pull the stretch-release tape strips out slowly and flat, parallel to the panel.
3. ☐ Tip the panel forward and disconnect the display cable.
4. ☐ Fit new tape (kit N13937-001) before mounting the new panel.
5. ☐ Connect the new panel and test the picture **before** fixing it down.
6. ☐ Refit everything in reverse.

Verify before closing the ticket

- Full image: no lines, dead pixels or uneven backlight.
- Brightness works across its whole range.
- Image steady through full lid travel.

R12 Display (eDP) cable replacement

LEVEL 2

Fixes	Flicker, lines or black screen that changes with lid angle; external display works
Part	Display cable N12926-001
Time	40–50 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape

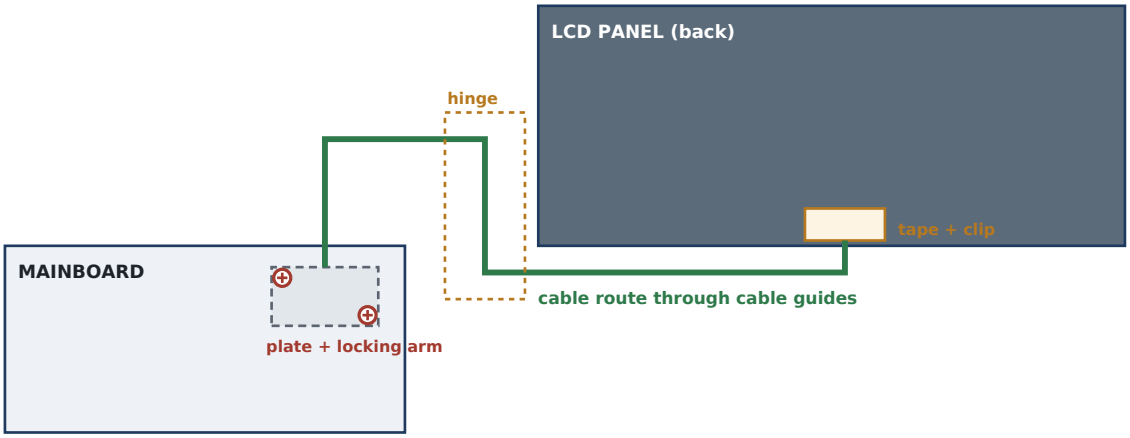


Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.

Procedure

1. ☐ Remove the bezel and panel; disconnect the display cable at the system board.
2. ☐ Free the cable from the back cover and hinge area; note every clip and tape point.
3. ☐ Route the new cable exactly along the old path and reconnect both ends.
4. ☐ Test at every lid angle before reassembly.

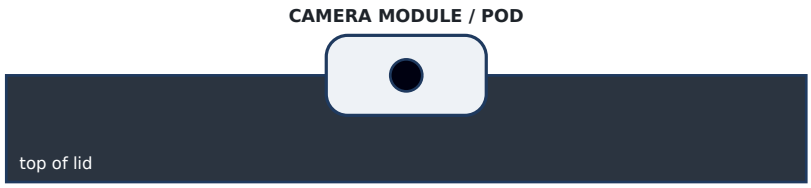
Verify before closing the ticket

- Steady image at every angle: open and close the lid 20 times while watching.
- Camera still works if it shares the cable.

R13 Camera module replacement

LEVEL 2

Fixes	Camera not detected, black image, broken lens
Part	Webcam module N12929-001; webcam cable N10434-001
Time	15–20 minutes
Tools	Plastic picks, plastic tweezers



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

Check first: privacy shutter open? Camera works after S1 and a powerwash (S3)?

Procedure

1. ☐ Remove the bezel.
2. ☐ Remove the 3 Phillips M2.0×2.0 webcam bracket screws.
3. ☐ Disconnect the webcam cable (N10434-001) and lift the module out.
4. ☐ Fit the new module, reconnect, and test before refitting.

Verify before closing the ticket

- Camera app shows a clear image.
- Privacy shutter fully covers the lens when closed.
- Works in a Google Meet test call.

R14 Hinge replacement

LEVEL 2

Fixes	Floppy lid, cracked hinge, grinding, lid won't stay up
Part	Hinges N12927-001; hinge cover N12928-001
Time	30–45 minutes
Tools	Phillips driver, plastic picks

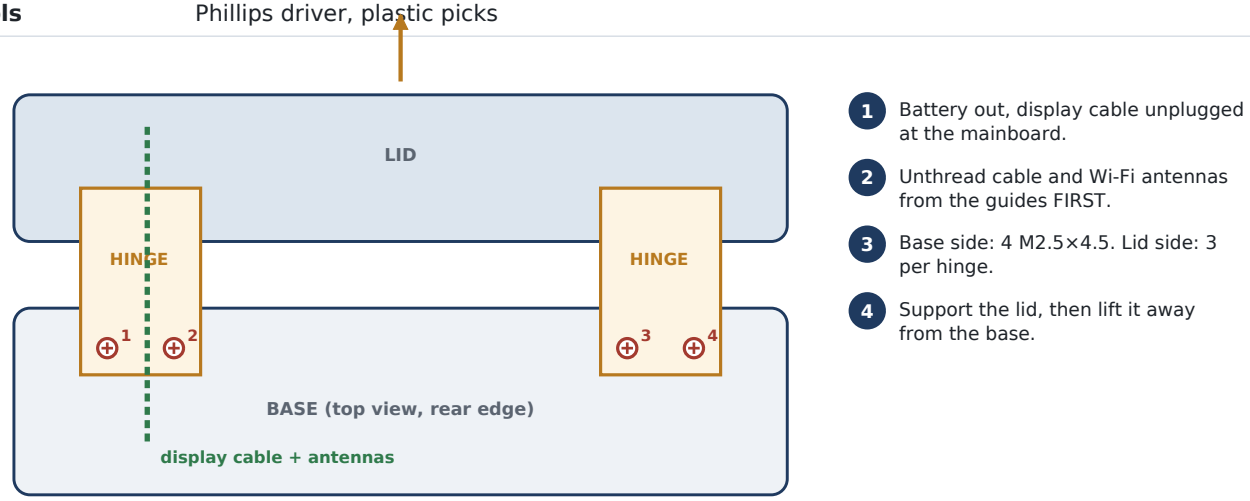


Illustration: hinge area seen from above with the D cover off. Positions approximate.

Check first: tighten the hinge screws. A screw that spins without tightening means a stripped post: that's a cover replacement.

Procedure

1. ☐ Remove the bottom cover, disconnect the battery, and separate the display assembly from the base.
2. ☐ Remove the hinge cover.
3. ☐ On each hinge remove 2 Phillips M2.5×2.5 broad-head screws and 1 Phillips M2.5×3.0 screw.
4. ☐ Lift the hinges off the lid.
5. ☐ Fit the new hinges (left and right differ) and reassemble.

Verify before closing the ticket

- Lid holds any angle through its full travel.
- Both hinges feel equally stiff.
- Display steady and Wi-Fi strength normal (no pinched cables).

R15 Keyboard / top cover replacement

LEVEL 3

Fixes	Several dead keys, a dead row, liquid-damaged keyboard, cracked palm rest
Part	Top cover with keyboard N14927-xxx (by country and backlight; look up by serial)
Time	60–90 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap, screw mat, Kapton tape

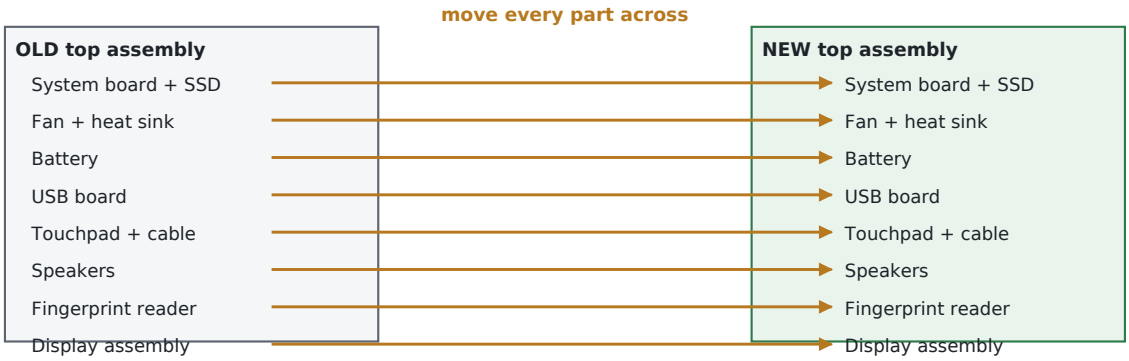


Illustration: what moves to the new top assembly. Check what comes pre-fitted first.

The keyboard is built into the top cover. HP's top cover "remains after all other spare parts are removed": this is a full transplant.

Procedure

1. ☐ Remove the bottom cover and the battery.
2. ☐ Remove every part listed below from the old top cover, in this order: battery, USB board, touchpad and cable, speakers, fingerprint reader and board, fan, heat sink, system board, display assembly.
3. ☐ Check what's pre-fitted on the new top cover, then transfer everything else.
4. ☐ Reconnect every cable, checking against your photos.
5. ☐ Reconnect the battery and test everything before refitting the bottom cover.

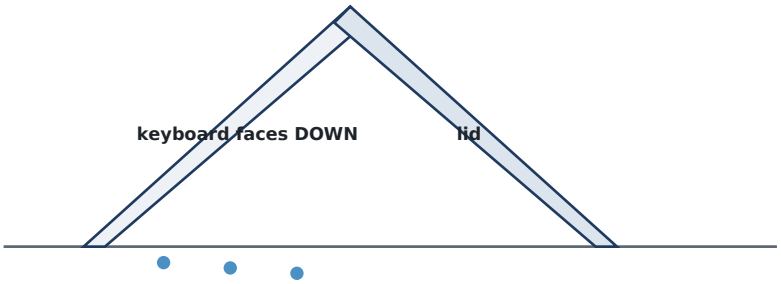
Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, ports, speakers and Wi-Fi all work.
- No screws left over.

R16 Liquid spill response

LEVEL 3

Fixes	Any liquid on or in the device
Part	Often none; sometimes keyboard/top cover or system board
Time	10 min first response; 24–48 h drying
Tools	90%+ isopropyl alcohol, soft brush, lint-free cloths



1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

Illustration: drain position after a spill.

Spill-resistant keyboards resist drips, not a full cup. Speed matters more than anything.

Do not power on to test. Current through liquid causes the corrosion that kills boards.

Procedure

1. ☐ Hold Power for 10 seconds to force off. Unplug.
2. ☐ Tent the device upside down (lid at 90°, keyboard down) to drain.
3. ☐ Open it and disconnect the battery as soon as possible.
4. ☐ Blot visible liquid; don't wipe it around.
5. ☐ Sugary or sticky spills: clean affected areas with a soft brush and isopropyl alcohol.
6. ☐ Dry open, battery disconnected, for 24 hours minimum (48 if soaked). No hair dryer, no rice.
7. ☐ Inspect for white or green corrosion; clean connectors.
8. ☐ Reconnect the battery, plug in, and run S2 fully.

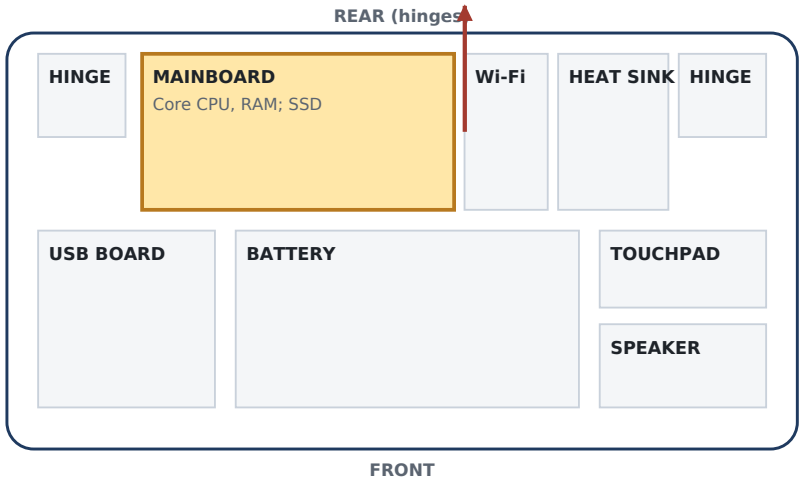
Verify before closing the ticket

- All S2 tests pass.
- Sticky or dead keys: keyboard repair.
- Corrosion on the system board: Section 6.

R17 System board replacement (lead tech)

LEVEL 3

Fixes	Dead USB-C ports, no power with a good battery, failed storage, RAM errors, failed recovery
Part	System board N13676-001 (by configuration; look up by serial)
Time	45–90 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap



1. Battery out first (and the Wi-Fi card if it's socketed).
2. Unplug: display, USB board, touchpad, keyboard, fingerprint, speaker, antennas
3. Lift out after the bracket and all cables are free.
4. Lift by the edges onto the ESD mat.

Illustration: base interior, mainboard highlighted. Red arrow = lift side.

Identity and enrollment. Storage, serial number and enrollment live on the system board. The lead tech updates the asset record, deprovisions the old device in the Admin console if needed, and plans re-enrollment. Ask the manufacturer whether the new board ships with the serial set.

Confirm first with S1, S2, S3 and a connector reseat. Most "dead boards" are a battery, cable or connector.

Procedure

1. ☐ Remove the bottom cover, the battery, the fan and heat sink.
2. ☐ Lift the Mylar and remove the bracket screws: 2 Phillips M2.0×3.0 and 1 Phillips M2.0×2.5.
3. ☐ Disconnect every cable from the board (photograph first).
4. ☐ Remove the 2 Phillips M2.0×2.5 board screws and lift the board out.
5. ☐ Move the SSD to the new board.
6. ☐ Fit the new board, reconnect every cable, refit the parts removed, then run recovery (S3) if needed and enroll.

Verify before closing the ticket

- Boots, enrolls and passes all of S2.
- Asset record and Admin console show the correct serial number.

6 Out of scope: escalate

These need soldering or board-level work, or parts that aren't sold. Tag the device with the ticket and send it to the lead tech.

Fault	Why	Route
USB-C port loose or dead	Soldered to the system board	System board or HP warranty
Storage or RAM failure	Soldered (or system-board level)	System board or HP warranty
Wi-Fi failure that S3 doesn't fix	No separate Wi-Fi spare listed by HP	HP warranty / system board
Burn marks, melted plastic, burnt smell	Fire hazard	Lead tech immediately
Any repair needing a soldering iron or hot air	Program policy	HP warranty or lead tech

HP support

- **HP support:** support.hp.com
- **HP parts lookup:** partsurfer.hp.com
- **Service guide:** HP Elite c640 14 inch G3 Chromebook Maintenance and Service Guide

7 Sources

Checked October 2026. The procedures are the guide author's adaptation of these sources for a student bench; they are not manufacturer-issued instructions.

- HP, *Maintenance and Service Guide: HP Elite c640 14 inch G3 Chromebook* (kaas.hpcloud.hp.com, pdf_6048310) — screw counts, sizes, part numbers and procedure order.
- HP PartSurfer (partsurfer.hp.com) — confirm part numbers by serial number.